

List of Publications

Burak Demirel

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Profile

I have conducted high-risk high-reward research in different institutions around the world and taught several courses in multicultural environments. I have published one book and over 25 papers, including peer-reviewed journal and conference publications (cited over 500 times).

Google Scholar Profile

Books

- [B1] L. Güvenç, B. A. Güvenç, B. Demirel and M. T. Emirler, *Control of Mechatronic Systems*, the IET Books, 216 pp, June 2017, Cloth: 978-1-78561-144-5.

Preprints

- [PP1] P. Soldati, B. Demirel, Y. A. Wang and M. Sintorn, “Artificial Intelligence in Intent Management for Differentiated Connectivity in Autonomous Networks”, *IEEE Communication Magazine, Artificial Intelligence and Data Science for Communications Series*, 2026.

Non-peer reviewed publications

- [TR3] P. Soldati, C. Tatino, B. Demirel, and J. Wszolek, “When AI Has No Time to Think: Inference Under Extreme Latency and Compute Constraints in RANs”, *Ericsson Technology Review*, no. 3, pp. 1-12, March 2026, (pdf).
- [TR2] B. Demirel, P. Soldati, and Y. A. Wang, “From Intents to Actions: Agentic AI in Autonomous Networks”, 2025, (arXiv:2602.01271).
- [TR1] P. Soldati, E. Ghadimi, B. Demirel, Y. Wang, M. Sintorn, and R. Gaigalas, “Approaching AI-Native Radio Access Networks through Generalization and Scalability of Learning”, *Ericsson Technology Review*, vol. 2023, no. 3, pp. 2-12, March 2023, doi: 10.23919/ETR.2023.10068317.

Journal publications

The first author (of the articles seen below) is the one who has contributed the most to the papers and has written almost the entire manuscripts.

- [J11] B. Demirel, Y. A. Wang, Cristian Tatino, and P. Soldati, “Generalization in Reinforcement Learning for Radio Access Networks”, *IEEE Transactions on Machine Learning in Communications and Networking*, 2026. arXiv:2507.06602
- [J10] P. Soldati, E. Ghadimi, B. Demirel, Y. Wang, R. Gaigalas, M. Sintorn, “Design Principles for Model Generalization and Scalable AI in Radio Access Networks”, *IEEE Communication Magazine, Artificial Intelligence and Data Science for Communications Series* vol. 63, no. 1, pp. 84-91, Jan. 2025. arXiv:2306.06251
- [J9] B. Demirel, A. S. Leong, V. Gupta and D. E. Quevedo, “Trade-Offs in Stochastic Event-Triggered Control”, *IEEE Transactions on Automatic Control*, vol. 64, no. 6, pp. 2567-2574, June 2019.
- [J8] B. Demirel, A. Ramaswamy, D. E. Quevedo and H. Karl, “DeepCAS: A Deep Reinforcement Learning Algorithm for Control-Aware Scheduling”, *IEEE Control Systems Letters*, vol. 2, no. 4, pp. 737-742, October 2018.
- [J7] B. Demirel, E. Ghadimi, D. E. Quevedo and M. Johansson, “Optimal Control of Linear Systems with Limited Control Actions: Threshold-Based Event-Triggered Control”, *IEEE Transactions on Control of Network Systems*, vol. 5, no. 3, pp. 1275-1286, September 2018.
- [J6] J. Wu, D. E. Quevedo, X. Cao, J. Xiong, Z. Ren and B. Demirel, “Guest Editorial Special Issue on Resource-Efficient Control in Cyber-Physical Systems”, *IET Control Theory & Applications*, vol. 11, no. 11, pp. 1663-1665, July 2017.
- [J5] B. Demirel, V. Gupta, D. E. Quevedo and M. Johansson, “On the Trade-Off Between Control Performance and Communication Cost in Dead-Beat Event-Triggered Control”, *IEEE Transactions on Automatic Control*, vol. 62, no. 6, pp. 2973 - 2980, June 2017.
- [J4] B. Demirel, Z. Zou, P. Soldati and M. Johansson, “Modular Design of Jointly Optimal Controllers and Forwarding Policies for Wireless Control”, *IEEE Transactions on Automatic Control, Special Issue on Control of Cyber-Physical Systems*, vol. 59, no. 12, pp. 3252 - 3265, Dec. 2014.
- [J3] B. Demirel, C. Briat and M. Johansson, “Deterministic and Stochastic Approaches to Supervisory Control Design for Networked Systems with Time-varying Communication Delays”, *Special Issue on the IFAC ADHS 2012 Conference in Nonlinear Analysis: Hybrid Systems*, vol. 10, pp. 94 - 110, 2013.

- [J2] B. Demirel, M. T. Emirler, U. Sönmez and A. Yörükoglu, “Semi-compliant Force Generator Mechanism Design for a Required Impact and Contact Forces”, *ASME Journal of Mechanisms and Robotics*, vol. 2, no. 4, Nov. 2010.
- [J1] B. Demirel and L. Güvenç, “Parameter Space Design of Repetitive Controllers for Satisfying a Robust Performance Requirement”, *IEEE Transactions on Automatic Control*, vol. 55, no. 8, pp. 1893 - 1899, Aug. 2010.

Book chapters

- [BC2] B. Demirel (Student Contributor) “Partially Compliant Force-Generator Mechanism”, Handbook of Compliant Mechanisms by L. L. Howell, S. P. Magleby and B. M. Olsen, Wiley, Feb. 2013.
- [BC1] B. Aksun Güvenç, S. Necipoglu, B. Demirel and L. Güvenç, “Robust Control of Atomic Force Microscopy”, Mechatronics by J. Paulo Davim, Wiley-ISTE, March 2011.

Conference publications

- [C15] S. Khosravi, B. Demirel, L. Zhou, J. Rasines and Pablo Soldati, “Practical Policy Distillation for Reinforcement Learning in Radio Access Networks”, *Proceedings of IEEE International Symposium on Personal, Indoor and Mobile Radio Communications*, Istanbul, Turkey, 2025.
- [C14] B. Demirel and A. Aytekin, “Application of Monte Carlo Tree Search in Periodic Schedule Design for Networked Control Systems”, *Proceedings of the European Control Conference (ECC’21)*, Rotterdam, the Netherlands, 2021.
- [C13] B. Demirel, A. Ramaswamy, D. E. Quevedo and H. Karl, “DeepCAS: A Deep Reinforcement Learning Algorithm for Control-Aware Scheduling”, *Proceedings of the 57th IEEE Conference on Decision and Control (CDC’18)*, 2018.
- [C12] B. Demirel, E. Ghadimi and D. E. Quevedo, “Sparsity-Promoting Norm-Optimal Iterative Learning Control for Resource-Constrained Control Systems”, *Proceedings of the 56th IEEE Conference on Decision and Control (CDC’17)*, 2017.
- [C11] B. Demirel, A. S. Leong and D. E. Quevedo, “Performance Analysis of Event-Triggered Control Systems with A Probabilistic Triggering Mechanism: The Scalar Case”, *Proceedings of the IFAC World Congress*, Toulouse, France, July 9-14, 2017.
- [C10] B. Demirel, V. Gupta, D. E. Quevedo and M. Johansson, “Threshold Optimization of Event-Triggered Multi-Loop Control Systems”, *Proceedings of the 13th International Workshop on Discrete Event Systems (WODES’16)*, Xi’an, China, May 30 - June 1, 2016.

- [C9] B. Demirel, H. R. Feyzmahdavian, E. Ghadimi and M. Johansson, “Stability Analysis of Discrete-Time Linear Systems with Unbounded Stochastic Delays”, *Proceedings of the 5th IFAC Workshop on Distributed Estimation and Control in Networked Systems (NECSYS’15)*, Philadelphia, Pennsylvania, Sept. 10 - 11, 2015.
- [C8] B. Demirel, A. Aytekin, D. E. Quevedo and M. Johansson, “To Wait or To Drop: On the Optimal Number of Re-transmissions in Wireless Control”, *Proceedings of the European Control Conference (ECC’15)*, Linz, Austria, July 15 - 17, 2015.
- [C7] B. Demirel, V. Gupta and M. Johansson, “On the Trade-Off Between Control Performance and Communication Cost for Event-Triggered Control over Lossy Networks”, *Proceedings of the Biannual European Control Conference (ECC’13)*, Zurich, Switzerland, July 17 - 19, 2013.
- [C6] Z. Zou, B. Demirel and M. Johansson, “Minimum-Energy Packet Forwarding Policies for LQG Performance in Wireless Control Systems”, *Proceedings of the 51st IEEE Conference on Decision and Control (CDC’12)*, Maui, Hawaii, Dec. 2012.
- [C5] B. Demirel, C. Briat and M. Johansson, “Supervisory Control Design for Networked Systems with Time-Varying Communication Delays”, *Proceedings of the 4th IFAC Conference on Analysis and Design of Hybrid Systems (ADHS’12)*, Eindhoven, the Netherlands, June 2012.
- [C4] B. Demirel, Z. Zou, P. Soldati and M. Johansson, “Modular Co-design of Controllers and Transmission Schedules in WirelessHART”, *Proceedings of the 50th IEEE Conference on Decision and Control and European Control Conference (CDC’11)*, Orlando, FL, USA, Dec. 2011.
- [C3] B. Demirel and L. Güvenç, “Control of Mechatronics Systems - COMES Toolbox”, *Proceedings of the ASME Biennial Conference on Engineering Systems Design and Analysis*, Istanbul, Turkey, July 12 - 14, 2010.
- [C2] S. Necipoglu, B. Demirel and L. Güvenç, “Fast AFM Scanning with Parameter Space Based Robust Repetitive Control Designed Using COMES Toolbox”, *Proceedings of the ASME Biennial Conference on Engineering Systems Design and Analysis*, Istanbul, Turkey, July 12 - 14, 2010.
- [C1] B. Demirel, M.T. Emirler, A. Yörükoglu, N. Koca and U. Sönmez, “Compliant Impact Generator for Required Impact and Contact Force”, *Proceedings of the ASME International Mechanical Engineering Congress and Exposition (IMECE’08)*, Boston, MA, Oct. 31 - Nov. 06, 2008.

Patents

- [P15] P. Soldati and **B. Demirel**, “Joint training of first and second AI models,” US Patent Application, 2025 (filed).

- [P14] P. Soldati and **B. Demirel**, “Method for distributed exploration for multi-objective RL in radio access networks,” US Patent Application, 2025 (filed).
- [P13] P. Soldati, E. Ghadimi, and **B. Demirel**, “Methods for AI-driven optimization of intents and resource utilization,” US Patent Application, 2023 (filed).
- [P12] P. Soldati, E. Ghadimi, and **B. Demirel**, “Methods for generalizing AI design in communication systems,” US Patent Application, 2023 (filed).
- [P11] H. S. Ghadikolaei, L. Eleftheriadis, R. Inam, **B. Demirel**, Z. Ghebretensaé, E. Sanders, “Management of delivery of power to a radio head,” WO2023085983, 2023.
- [P10] H. S. Ghadikolaei, L. Eleftheriadis, **B. Demirel**, A. Azari, Z. Ghebretensaé, “Allocating computing tasks to radio network nodes,” WO2023078547, 2023.
- [P9] P. Soldati, E. Ghadimi, Yu Wang, and **B. Demirel**, “Inter-node exchange of link adaptation assistance information,” WO2022229181, 2022.
- [P8] P. Soldati, E. Ghadimi, Yu Wang, and **B. Demirel**, “Method for transmitting link adaptation state information in telecommunication networks,” WO2022162152, 2022.
- [P7] P. Soldati, E. Ghadimi, Y. Wang, **B. Demirel**, and H. Rydén, “Method for user device assisted link adaptation in communication networks,” WO2022162148, 2022.
- [P6] J. Jeong, **B. Demirel**, H. Tated, and W. Hu, “Verifying an action proposed by a reinforcement learning model,” WO2022248040, 2021.
- [P5] K. Cyras, M. Daoutis, A. Pradeep Mujumdar, A. Kattepur, and **B. Demirel**, “Determining whether a device can perform an action,” WO2022242826, 2021.
- [P4] K. Vandikas, A. Vulgarakis Feljan, **B. Demirel**, M. Orlic, A. Previti, and A. Nikou, “Reinforced federated learning utilizing multiple specialized machine learning agents,” WO2022045937, 2020.
- [P3] S. Içkin, T. Kvernvik, R. Inam, and **B. Demirel**, “Controlling communications from content server to end device”, WO2021239250, 2020.
- [P2] **B. Demirel**, P. Valenzua, P. Lima, and T. Nyberg, “Method and control arrangement for controlling the movement of an autonomous vehicle”, DE102020001887, 2020.
- [P1] **B. Demirel**, P. G. Collares, H. Pettersson, M. Granström, and P. Lima, “Method and control unit for determining an oscillatory behavior of a vehicle”, SE1950874, 2019.

Theses

- [T3] B. Demirel, “Architectures and Performance Analysis of Wireless Control Systems”, Doctoral Thesis, KTH Royal Institute of Technology, Stockholm, Sweden, May 2015. ISBN 978-91-7595-528-5. **The material presented in this thesis relies mainly on [J3-J5, C4-C8].**
- [T2] B. Demirel, “Design and Performance Analysis of Wireless Networked Control System”, Licentiate Thesis, KTH Royal Institute of Technology, Stockholm, Sweden, May 2013. ISBN 978-91-7501-748-8. **The material presented in this thesis relies mainly on [J3, C4-C7].**
- [T1] B. Demirel, “Interactive computer-aided controller design for mechatronic systems”, Master of Science Thesis, Istanbul Technical University, Istanbul, Turkey, September 2009. **The material presented in this thesis is based on [J1, C2, C3].**